ARTIFICIAL INTELIGENCE - CSA1709

Assessment tool - 3

R. Thilagavathi
1972260045

Q.1 INTELLIGENT AGENTS IN SMART HOME AUTOMATION:

The Intelligent agents involving in smart home activities including lights, AC, temperature, security and others are of four types, which are listed below;

1) Simple Reflex Agent
2) Model - Based Agent
3) Goal Based Agent
4) Utility Based Agent

1) SIMPLE REFLEX AGENT:

This simple reflex agent takes action based on the current situation.

Example: If motion is detected, the light turns ON. If no motion is detected, the light turns OFF.

Advantages: It is fast and simple

Disadvantages: It cannot remember past situations

2) MODEL BASED AGENT:

A model Based agent uses current information and remembers previous situations to make decisions.

Example: It remembers the room temperature and adjust the AC when needed.

3) DIFFERENCE BETWEEN UCS AND IDS :

USC selects the path with the lowest total cost, whereas IDS searches by increasing the depth step by step. UCS requires more memory, while IDS uses less memory. UCS is suitable when path costs are important, whereas IDS is useful when memory is limited.

4) TYPES OF INTELLIGENT AGENTS:

SIMPLE REFLEX AGENT: It takes action based on the current situation.

MODEL-BASED AGENT: It remembers information about the maze and obstacles.

GOAL-BASED AGENTS: It chooses action to reach the exit.

UTILITY BASED AGENTS: It chooses the best action based on it benefits.

5) BEST AGENT AND SEARCH STRATEGY:

A model Based Agent combined with IDS is suitable when the robot has limited memory. The model-Based agent remembers the explored paths and obstacles while IDS searches step by step.

However, if finding the lowest-cost path is important, UCS is a better choice.

Q.2 ROBOT FINDING AN EXIT IN AN UNKNOWN MAZE:

INTRODUCTION:

A Robot placed in an unknown maze must find the exit without knowing the paths. search algorithms such as Uniform Cost search (UCS) and Iterative Deepening Search (IDS) help the robot find the exit.

1) UNIFORM COST SEARCH (UCS)

Uniform Cost search selects the path with the lowest total cost first.

Example: If two paths have different costs, the robot explores the cheaper path first.

Advantages: It finds the lowest cost path and gives an optimal solution when the required conditions are satisfied.

Disadvantages: It requires more memory and may take more time in a large maze.

2) ITERATIVE DEEPENING SEARCH (IDS):

Iterative Deepening search searches the maze by increasing the depth step by step. It first searches at depth 0, then depth 1, depth 2 and continues until it finds the exit

Advantages: It uses less memory and find the exit when the search space is finite.

Disadvantage: It Repeats some searches and it does not always find the lowest cost path.

Advantages: It can make better decisions using stored informations.

Disadvantages: It requires more memory.

Q.2

3) GOAL BASED AGENT :

A goal Based agent takes actions to achieve a specific goal.

Example : It controls the AC to maintain the desired room temperature.

Advantages: It works towards a specific goal.

Disadvantages: It may take more time to decide the best actions.

4) UTILITY BASED AGENT :

A Utility Based agent selects the best action by comparing different options.

Example : It maintains a comfortable temperature while reducing electricity usage.

Advantage : It balances comfort and energy efficiency.

Disadvantage : It can be complex to design.

BEST AGENT:

A hybrid approach combining all four agents is the best choice for a smart home. The simple reflex agent responds quickly, the model Based agents remembers information, the goal-based agents achieves targets and the utility based agent saves energy.